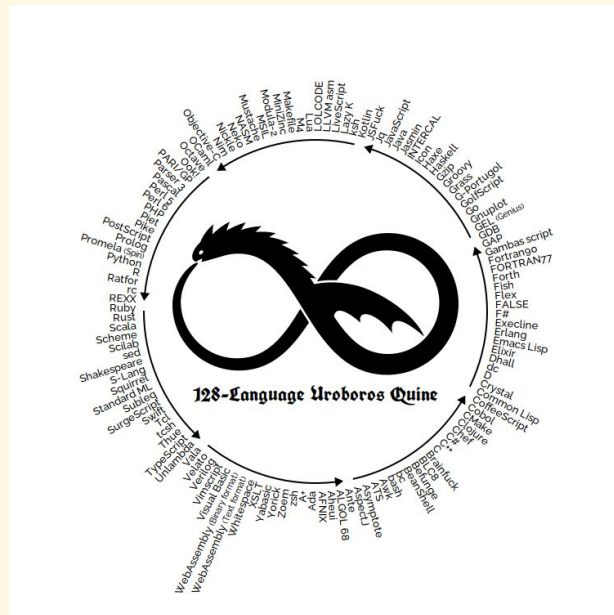


Hello!

👋 I'm Noon



Packaging 128 programming languages with Nix



What is a programming language?

~~What is a programming language?~~

What is a quine?

A program p , with source code s , that, when evaluated, outputs s .

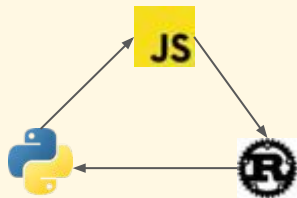
Example (Haskell):

Further reading: github.com/kirjavascript/quine-howto
“How to write lots of quines”

```
import Text.Printf (printf)
main =
  let s = "import Text.Printf (printf)\nmain =\n  let s = %s\n  in printf s (show s)\n"
  in printf s (show s)
```

Then a quine relay is ...

A program **p**, with source code **s**, that, when evaluated, outputs a program **q**, ..., that when evaluated produces **s**!



Some natural questions may arise ...

“Is my programming language there?”

“How can I add a new language!?”

“I want to see the source code for language ...”

Some of these questions involve running it yourself

github.com/mame/quine-relay

```
bash QR.bash > QR.bc

#####
## 43: bc → BeanShell ##
#####

BC_LINE_LENGTH=4000000 bc -q QR.bc > QR.bsh

#####
## 44: BeanShell → Befunge ##
#####

bsh QR.bsh > QR.bef
stty: Inappropriate ioctl for device
make: *** [Makefile:401: QR.bef] Error 1
make: *** Deleting file 'QR.bef'
```

Docker instructions fail at the moment.



Question: Have you ever seen
readme installation instructions fail?

What if it didn't have to be that way ...



Nix: What is it?



Question: What have you heard?

- Package manager?
- Build tool?
- Programming language?
- Operating system?
- ... *Restaurant in Portobello?*



Nix: A quick look 🧐

- Package manager
 - `nix-shell -p ...`
 - search.nixos.org
 - “[nixpkgs](https://search.nixos.org/packages)”
- Programming language?
 - `let f = x: x + 2;`
`in f 5;`
 - By far the least interesting fact of Nix
- Operating system?
 - Kinda; it's still linux



Nix: Still looking quickly ... 🧐

- Packaging
 - *<Quick demo ...>*
- “devShells”
 - *<Quick demo ...>*



Nix: Brief theory overview

- Derivations
 - `pkgs.stdenv.mkDerivation { ... }`
 - *<Example on nixpkgs>*
- Nix store
 - `/nix/store/...`
- Build sandboxing
 - No connecting to the internet (without “FOD”)
 - => Always reproducible
- Isolation/Content-based hashing
 - “No more dependency hell”
 - Immediate rebuilds
 - “No more docker hell”
 - *<Example>*



Okay. So the challenge is clear.

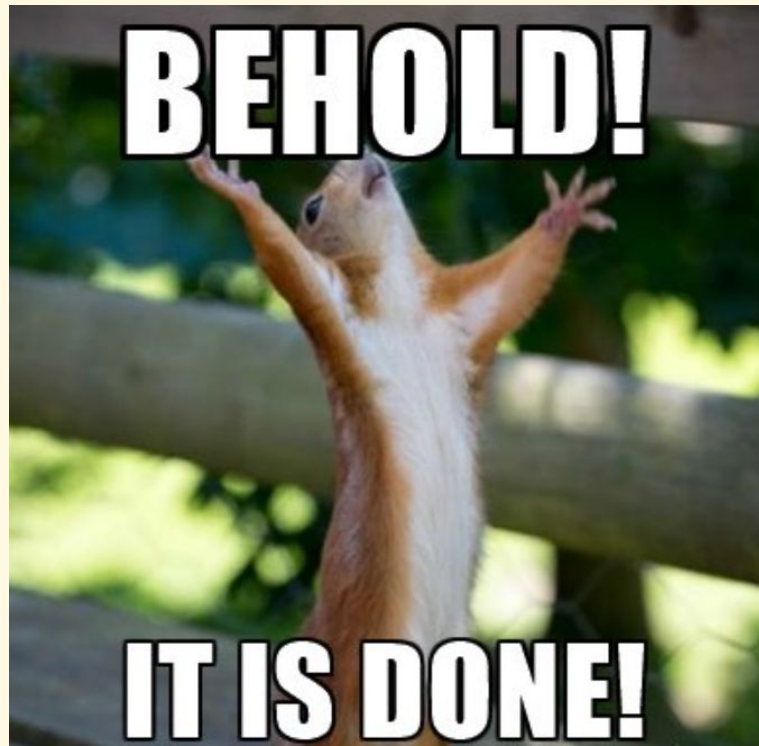
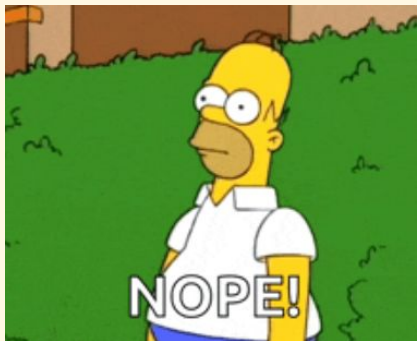
- Are all the languages of quine-relay in nixpkgs?
 - 95 yes; 33 “no”.
- What of the “no” packages?
 - Almost all are “simple” to build
 - Some had to be “patched”
 - Only two were “problematic”
 - **<Let’s take a look>**
- Gambas3
 - Requires a “fake” FHS env to run: /usr/lib ...
 - Easy to patch
- Modula-2
 - Inside the giant and complex “gcc” monorepo
 - Luckily, can patch nixpkgs gcc to build it!



We need packages ... lots of packages.

Conclusion

- Let's take a look ..
 - *<Demo ...>*
- Read more:
 - invariant.club/articles.html
- Any downsides?



Thanks!

<https://invariant.club>

Most relevant ▾



Rok Garbas ✓ • 1st

Engineer at Flox | DevOps/SRE/Release Eng | Cust...

19h ...

"only" 128 languages? 😊

Funny · 🗨️ 1 | Reply

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Bonus content: Adding new languages

<https://github.com/silky/quine-relay/pull/2>